

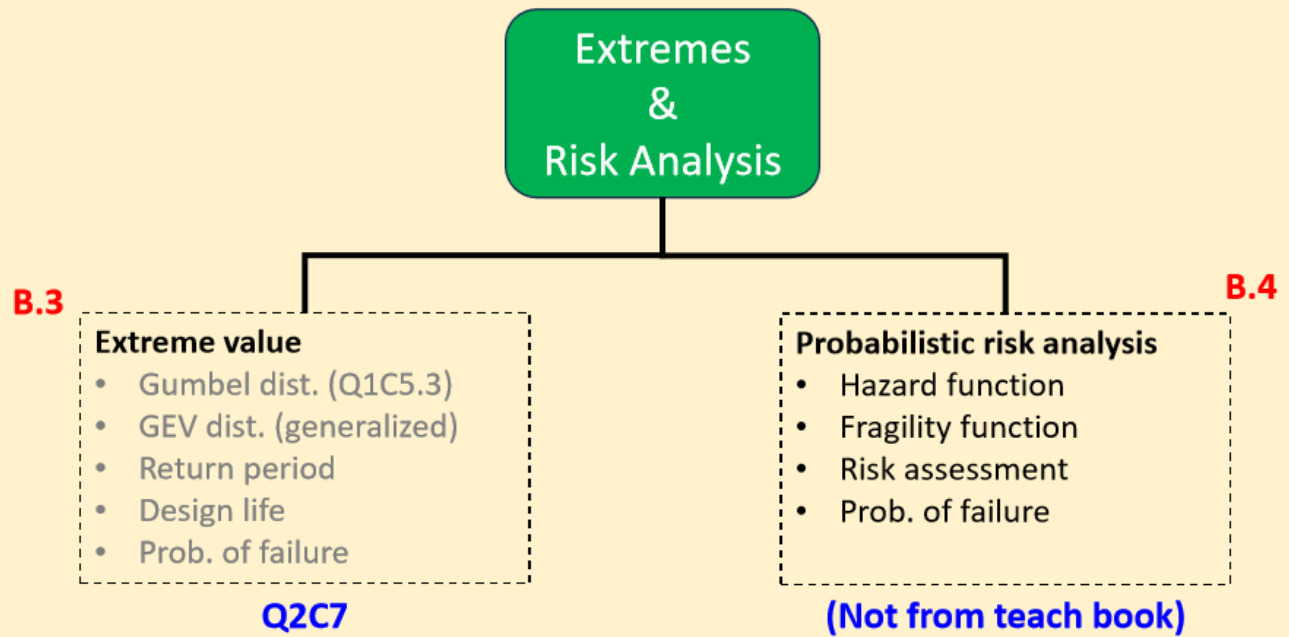
CV 5101

Modeling, Uncertainty, and Data for Engineers

(July – Nov 2026)

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Module Overview



Flow

- Risk, reliability, resilience
- Typical probabilities
- Formalizing monkey attack
- Probabilistic monkey attack risk
- Probabilistic seismic risk

Risk, reliability, resilience

- p_f : Probability of Failure
- Risk: Expected loss
- Reliability: $P_r(\text{system performs its function without failure}) \approx 1 - p_f$
- Resilience: Ability of a system to withstand, recover, restore functionality after disruption.

Typical Probabilities

- Some **target probabilities** are:

- Locally dramatic but rather harmless in a wider area

e.g., building collapse

$\sim 10^{-4}$ for the lifetime of the facility

- More dramatic failure consequences, such as nuclear power plants

$\leq 10^{-5}$ /year core meltdown

$\leq 10^{-6}$ /year large leakage

- Drawing an ace of spade in a randomized deck of cards

$1/52 \approx 20 \times 10^{-3}$

- Rule of thumb,

probability of death/person/year $> 10^{-3} \Rightarrow$ Immediate Action

probability of death/person/year $< 10^{-6} \Rightarrow$ Event so unusual that

Nothing reasonable can be done.

Typical Probabilities

- Rule of thumb, **Probability of death/person/year**
 - $> 10^{-3} \Rightarrow$ Immediate Action
 - $< 10^{-6} \Rightarrow$ events so unusual that little can reasonably be done
 - Examples:
 - Worker in unregulated mine in early 20th century
often $> 10^{-2}$ / worker / year
 - Riding a motorbike without a helmet (in low-income countries)
 10^{-4}
 - Riding a motorbike with a helmet (in low-income countries)
 10^{-5}
 - Structural collapse of old and ill-maintained buildings
 - Commercial plane *1.2×10^{-8} / person-hour*
 - Car *0.7×10^{-8} / person-hour*
- Accounting for exposure*
- 24×10^{-6}*
 200×10^{-6}
- 6*

The R-S Reliability Problem

The R-S reliability problem

- Let R = capacity of a structure

S = demand on the structure

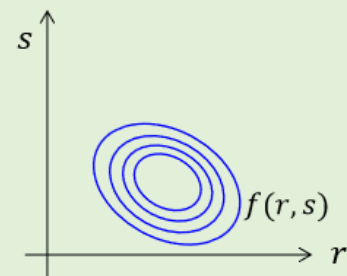
$f_{RS}(r, s)$ = joint PDF

Failure =

$p_f =$

$1 - p_f =$

$p_f =$



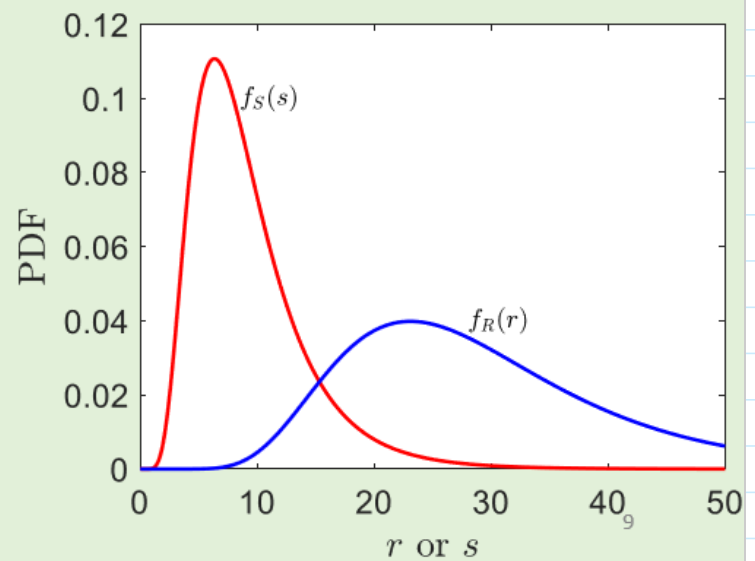
The R-S reliability problem

- Solution by conditioning on S

$$f_{RS}(r, s) =$$

$$p_f = \Pr(R \leq S) =$$

If R and S are statistically independent,



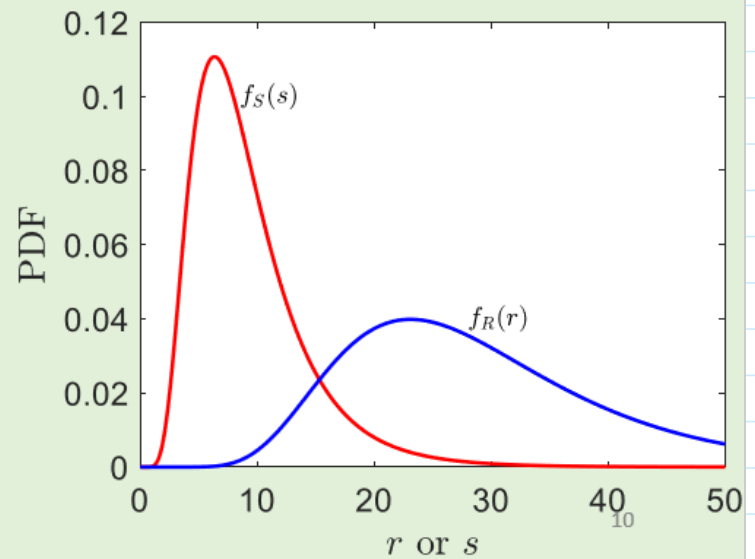
The R-S reliability problem

- Solution by conditioning on R

$$f_{RS}(r, s) =$$

$$p_f = \Pr(R \leq S) =$$

If R and S are statistically independent,



Skip safety margin, safety
factor of R-S problem,
if time does not permit

The R-S reliability problem

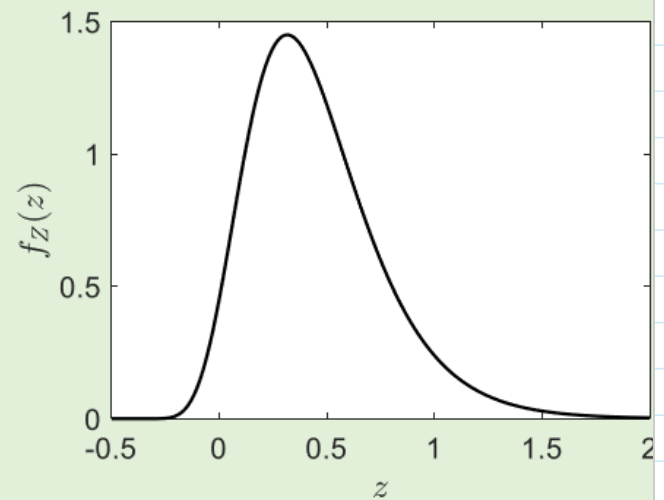
- Formulation in terms of safety margin

$$Z =$$

$$p_f =$$

$$\mu_Z =$$

$$\sigma_Z^2 =$$



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The R-S reliability problem

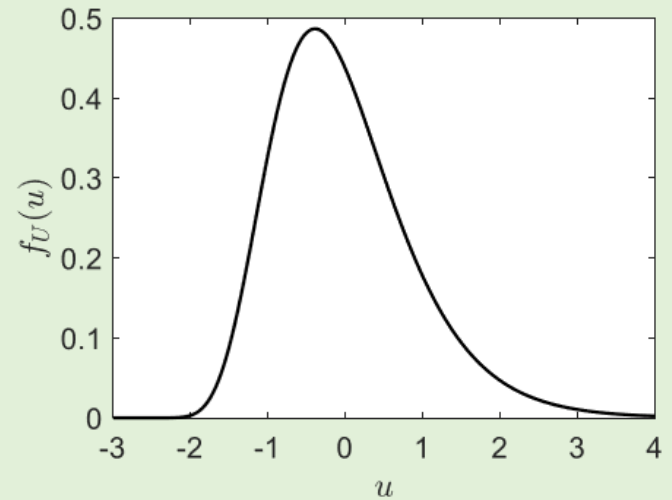
- Formulation in terms of safety margin

$$Z = R - S$$

$$p_f = \Pr(Z \leq 0) = F_Z(0)$$

$$U =$$

$$p_f =$$



The R-S reliability problem

- Special case: R and S are jointly normal (ok if small c.o.v.)

$$Z = N(\mu_Z, \sigma_Z), \quad U = N(0,1)$$

$$p_f = \Pr(Z \leq 0) = F_U(-\beta) = \Phi(-\beta)$$

$$\beta =$$

The R-S reliability problem

- Formulation in terms of (logarithmic) safety factor

$$Z = \ln R - \ln S, \quad U = \frac{Z - \mu_Z}{\sigma_Z}$$

Use the SO approximation for mean and the FO for Var.,

$$\mu_Z \cong$$

$$\sigma_Z^2 \cong$$

Recall, the SO approximation for mean,

$$\mu_{Y_k} \cong g_k(\mathbf{M}_X) + \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \left(\frac{\partial^2 g_k}{\partial X_i \partial X_j} \right)_{\mathbf{X}=\mathbf{M}_X} \rho_{X_i X_j} \sigma_{X_i} \sigma_{X_j}$$

Recall, the FO approximation for Var.,

$$\sigma_{Y_k}^2 \cong \nabla_{\mathbf{x}g_k}(\mathbf{M}_X) \Sigma_{\mathbf{X}\mathbf{X}} \nabla_{\mathbf{x}g_k}(\mathbf{M}_X)^T, \text{ where } \nabla_{\mathbf{x}g_k} = [\partial g_k / \partial X_1 \cdots \partial g_k / \partial X_n]$$

Note,

$$\left. \frac{\partial Z}{\partial R} \right|_{Z=\mu_Z} = \left. \frac{1}{R} \right|_{R=\mu_R} = \frac{1}{\mu_R}$$

$$\left. \frac{\partial Z}{\partial S} \right|_{Z=\mu_Z} = - \left. \frac{1}{S} \right|_{S=\mu_S} = - \frac{1}{\mu_S}$$

$$\left. \frac{\partial^2 Z}{\partial R^2} \right|_{Z=\mu_Z} = - \left. \frac{1}{R^2} \right|_{R=\mu_R} = - \frac{1}{\mu_R^2}$$

$$\left. \frac{\partial^2 Z}{\partial S^2} \right|_{Z=\mu_Z} = \left. \frac{1}{S^2} \right|_{S=\mu_S} = \frac{1}{\mu_S^2}$$

$$\frac{\partial^2 Z}{\partial R \partial S} = 0$$

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The R-S reliability problem

- Formulation in terms of (logarithmic) safety factor

$$Z = \ln R - \ln S$$

$$\mu_Z \cong \ln \mu_R - \ln \mu_S - \frac{1}{2}(\delta_R^2 - \delta_S^2)$$

$$\sigma_Z^2 \cong \delta_R^2 - 2\rho_{RS}\delta_R\delta_S + \delta_S^2$$

$$\beta \cong$$

The R-S reliability problem

- Special case: R and S are jointly lognormal

$$\mu_Z = \ln \left(r \sqrt{\frac{1+\delta_S^2}{1+\delta_R^2}} \right),$$

$$\sigma_Z^2 = \ln \left[\frac{(1 + \delta_R^2)(1 + \delta_S^2)}{(1 + \rho_{RS}\delta_R\delta_S)^2} \right]$$

$$\beta = \frac{\mu_Z}{\sigma_Z} =$$

The R-S reliability problem

- Reliability problem. **failure** = $R \leq S$
 - $p_f = \Pr(R \leq S)$. Hence, Reliability = $1 - p_f$
- Solution by **conditioning on S**. Sweep s from 0 to ∞ and find p_f .
- Solution by **conditioning on R**. Sweep r from 0 to ∞ and find p_f .
- Formulation in terms of **safety margin** $Z = R - S$, $U = \frac{Z - \mu_Z}{\sigma_Z}$
 - General definition, $\beta = \mu_Z / \sigma_Z$.
 - An approximation is to assume $U = N(0,1)$. Then $p_f \cong \Phi(-\beta)$
 - Special case: R and S are jointly normal. $\beta = \frac{\ln r - \frac{1}{2}(\delta_R^2 - \delta_S^2)}{\sqrt{\delta_R^2 - 2\rho_{RS}\delta_R\delta_S + \delta_S^2}}$, $p_f = \Phi(-\beta)$
- Formulation in terms of **safety factor** $Z = \ln R - \ln S$, $U = \frac{Z - \mu_Z}{\sigma_Z}$
 - SO approx. $\beta \cong \frac{\ln r - \frac{1}{2}(\delta_R^2 - \delta_S^2)}{\sqrt{\delta_R^2 - 2\rho_{RS}\delta_R\delta_S + \delta_S^2}}$ (Approximation but general).
 - An approx. is to assume $U = N(0,1) \Rightarrow p_f \cong \Phi(-\beta)$

$$\beta = \frac{\ln\left(r \sqrt{\frac{1+\delta_S^2}{1+\delta_R^2}}\right)}{\sqrt{\ln\left[\frac{(1+\delta_R^2)(1+\delta_S^2)}{(1+\rho_{RS}\delta_R\delta_S)^2}\right]}}$$
 - Special case: R and S are jointly lognormal. $\beta = \frac{\ln\left(r \sqrt{\frac{1+\delta_S^2}{1+\delta_R^2}}\right)}{\sqrt{\ln\left[\frac{(1+\delta_R^2)(1+\delta_S^2)}{(1+\rho_{RS}\delta_R\delta_S)^2}\right]}}$ (Exact but special)